

B.Tech, 4th SEMESTER
ESE EXAMINATION AIML (BLENDED), EVEN SEM. - 2025-26
AI Driven Full Stack Development (AI308B)

Time: 3 Hours

Total Marks: 50

Note: The project report in PDF format must be uploaded on Moodle during the examination. Part D

CASE STUDY: AI-Based Smart Complaint Management System

Design and implement a MERN Stack application that allows users to register complaints online. The system should use AI APIs to classify complaint priority, generate automated responses, and recommend the concerned department.

The application should support:

- Complaint registration
- Complaint tracking
- AI-based complaint categorization
- Secure authentication
- Deployment on Render.

Q1	All parts are compulsory	CO-3	
	FRONTEND REQUIREMENTS	BL	Marks
	Create the following modules: 1. Complaint Registration Form 2. Complaint List Page 3. Complaint Status Update Page 4. AI Analysis Result Display Complaint Registration Users can submit complaints with: • Name • Email • Complaint Title • Complaint Description • Complaint Category • Location • Complaint Status	3	8

	Complaint Tracking Users/Admin can: • View all complaints • Filter complaints by category • Update complaint status • Search complaints by location AI-Based Complaint Analysis Integrate AI API to: • Detect complaint urgency • Suggest responsible department • Generate automatic response message • Summarize complain		
Q2	All parts are compulsory	CO-2	
	BACKEND REQUIREMENTS (Node.js & Express.js)	BL	Marks
	Implement: 1. RESTful APIs 2. Controllers & Routes 3. Middleware 4. Validation & Error Handling Required API Endpoints Complaint APIs a). Add Complaint POST /api/complaints Sample Request Body <pre>{ "name": "Rahul Kumar", "email": "rahul@gmail.com", "title": "Water Leakage Issue", "description": "Water pipeline damaged near market area.", "category": "Water Supply", "location": "Ghaziabad" }</pre>	3	8
	b). Get All Complaints GET /api/complaints		

	3. Update Complaint Status PUT /api/complaints/:id												
	4. Search Complaint by Location GET /api/complaints/search?location=Ghaziabad												
	AI APIs 5. AI Complaint Analyzer POST /api/ai/analyze												
Q3	All parts are compulsory	CO-2											
	MongoDB Schema	BL	Marks										
	Schema Design 1. CRUD Operations 2. Query Filtering 3. Validation Test Cases <table><thead><tr><th>Test Case</th><th>Expected Output</th></tr></thead><tbody><tr><td>Add valid complaint</td><td>Complaint stored successfully</td></tr><tr><td>Missing title field</td><td>Validation error</td></tr><tr><td>Invalid email</td><td>Error message</td></tr><tr><td>Filter by location</td><td>Matching complaints displayed</td></tr></tbody></table> const ComplaintSchema = new mongoose.Schema({ name: String, email: String, title: String, description: String, category: String, location: String, status: { type: String, default: "Pending" }, createdAt: { type: Date, default: Date.now } });	Test Case	Expected Output	Add valid complaint	Complaint stored successfully	Missing title field	Validation error	Invalid email	Error message	Filter by location	Matching complaints displayed	3	6
Test Case	Expected Output												
Add valid complaint	Complaint stored successfully												
Missing title field	Validation error												
Invalid email	Error message												
Filter by location	Matching complaints displayed												

Q4	All parts are compulsory	CO-4											
	MERN INTEGRATION (Frontend–Backend–Database)	BL	Marks										
	Integrate: - React Frontend - Express Backend - MongoDB Database Test Cases<table><tr><th>Action</th><th>Expected Result</th></tr><tr><td>Submit complaint from frontend</td><td>Data saved successfully</td></tr><tr><td>Fetch complaints</td><td>Complaints displayed</td></tr><tr><td>Update complaint status</td><td>Updated status shown</td></tr><tr><td>Delete complaint</td><td>Complaint removed</td></tr></table>	Action	Expected Result	Submit complaint from frontend	Data saved successfully	Fetch complaints	Complaints displayed	Update complaint status	Updated status shown	Delete complaint	Complaint removed	3	6
Action	Expected Result												
Submit complaint from frontend	Data saved successfully												
Fetch complaints	Complaints displayed												
Update complaint status	Updated status shown												
Delete complaint	Complaint removed												
Q5	All parts are compulsory	CO-5											
	AI INTEGRATION	BL	Marks										
	AI Features - Complaint Priority Detection - Department Recommendation - Complaint Summary - Auto-generated User Response Test Cases<table><tr><th>Complaint Type</th><th>Expected AI Output</th></tr><tr><td>Water leakage</td><td>Water department suggestion</td></tr><tr><td>Electricity issue</td><td>High priority alert</td></tr><tr><td>Garbage complaint</td><td>Sanitation department</td></tr><tr><td>Long complaint text</td><td>AI-generated summary</td></tr></table>	Complaint Type	Expected AI Output	Water leakage	Water department suggestion	Electricity issue	High priority alert	Garbage complaint	Sanitation department	Long complaint text	AI-generated summary	3	8
Complaint Type	Expected AI Output												
Water leakage	Water department suggestion												
Electricity issue	High priority alert												
Garbage complaint	Sanitation department												
Long complaint text	AI-generated summary												

Q6	All parts are compulsory	CO-3											
	AUTHENTICATION & SECURITY	BL	Marks										
	Implement: 1. JWT Authentication 2. bcrypt Password Hashing 3. Login & Signup 4. Protected Routes Test Cases<table><tr><th>Test Case</th><th>Expected Output</th></tr><tr><td>Valid login</td><td>Token generated</td></tr><tr><td>Invalid password</td><td>Unauthorized error</td></tr><tr><td>Access without token</td><td>Access denied</td></tr><tr><td>Stored password</td><td>Encrypted format</td></tr></table>	Test Case	Expected Output	Valid login	Token generated	Invalid password	Unauthorized error	Access without token	Access denied	Stored password	Encrypted format	3	5
Test Case	Expected Output												
Valid login	Token generated												
Invalid password	Unauthorized error												
Access without token	Access denied												
Stored password	Encrypted format												
Q7	All parts are compulsory	CO-1											
	GIT & GITHUB USAGE	BL	Marks										
	Requirements 1. Proper Git Commits 2. GitHub Repository Upload 3. README.md File	3	3										
Q8	All parts are compulsory	CO-4											
	DEPLOYMENT ON RENDER	BL	Marks										
	Deploy: • Frontend • Backend • MongoDB Connection Submission Requirements • Live Frontend URL • Backend API URL	3	3										

	GitHub Repository Link		
Q9	All parts are compulsory	CO-1	
	CODE QUALITY & DOCUMENTATION	BL	Marks
	Evaluation Criteria 1. Folder Structure 2. Naming Conventions 3. Comments & Documentation 4. Reusable Components	3	3

NOTE: Make Docs and convert it into PDF for submission and it should contain:

- o Code
- o Code output (screenshots)
- o Postman/Thunder client all HTTP request screenshots
- o MongoDB items storage part (screenshots)
- o Render deployment successful (screenshot)
- o Render testing live URL for each end-point